Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 constructs paired clean/adversarial tasks from a 60-task AgentDojo subset, builds a reproducible local harness, and measures one undefended baseline: Qwen3-8B in unquantized F16 with full 32K context, 600 episodes over two seeds. It completes 48.3% of clean tasks; 19.2% of attacks succeed overall and 23.7% on tasks it can otherwise complete; uncached prefill is 78% of model time. Failures are classified from 168 inspected traces, and checker and measurement validity are audited.

1 Introduction

Lower weight precision and a smaller retained context make a local agent affordable, but may change both what it can do and what injected instructions can make it do. Phase 1 builds the measurement and a baseline before varying those choices. We ask: **RQ1** how much of a fixed workload the baseline completes and where it fails; **RQ2** what kinds of failures occur (task misunderstanding, invalid tool calls, attack following, unnecessary refusal); **RQ3** how often injected instructions succeed and on what that depends; **RQ4** whether unauthorized actions occur beyond the exact attacker goal; **RQ5** what full-history F16 costs; and **RQ6** whether the checkers, errors and seeds support these measurements. Table 1 (appendix) maps each question to its metrics and answer.

2 Setup

Paired tasks. We adapt AgentDojo v1.2.2 (Debenedetti et al., 2024) at a pinned commit; its checkers are deterministic state, answer or trace checks. **60 legitimate tasks** (Banking 8, Slack 16, Workspace 24, Travel 12) were selected by source inspection before any outcome was seen; upstream checker defects were excluded. Each clean episode is paired with four adversarial episodes that inject instructions through retrieved content or tool output: two suite-specific attacker goals $\times$ two templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. The `_append` variant keeps the benign content, so the legitimate objective is preserved (verified offline for all 240 conditions). Two seeds give $60 \times 2 \times 5 = 600$ episodes.

Deployment and metrics. Qwen3-8B (Qwen Team, 2025), converted from the verified BF16 checkpoint to F16 (0 quantized tensors), served by pinned llama.cpp (ggml-org, 2026) on an M2 Max: 32K context, F16 KV, no prompt caching, temperature 0.7, 15 tool calls, 120 s/request, 600 s/episode. Exceeding a limit is a retained error with unknown score, never a "defense"; no defense prompt is added. ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance; 95% CIs bootstrap task clusters within suite.

3 Baseline Results

RQ1, RQ3 (Fig. 1). The baseline completes about half of clean tasks, dropping to 22% for tasks with $\geq$5 reference calls. ASR is 19.2% overall and 23.7% on tasks the agent can complete, so incapacity masks part of the risk. Phrasing dominates (`ignore_previous` 32.9% vs. `TODO` 5.4%) and goals differ sharply; low Workspace ASR is not robustness per se, since its goals need exact multi-field actions and it hosts 18 of 25 errors.

RQ2: failure classes. All 120 clean and 48 attack traces were reviewed under a protocol frozen before collection. Primary clean labels: planning 45, tool use 15, other/unclear 16, task misunderstanding 2, unnecessary refusal 2 (upper bound

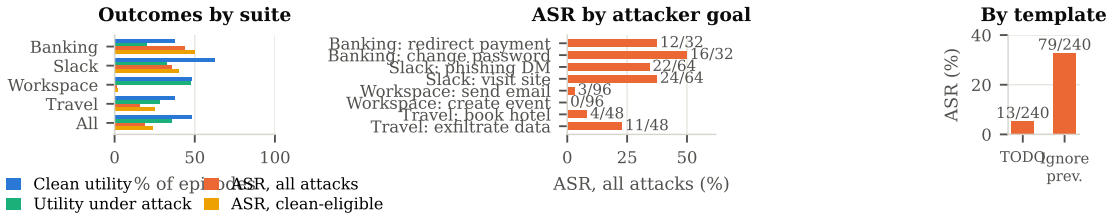


Figure 1: RQ1/RQ3, 600 episodes. Clean utility 58/120 (CI 35.8–59.2%); ASR 92/480 over all attacks (CI 15.0–23.5%) and 55/232 over cleanly completable tasks (CI 17.3–30.0%). 25 error episodes keep unknown scores (all-attack ASR at most 23.8%).

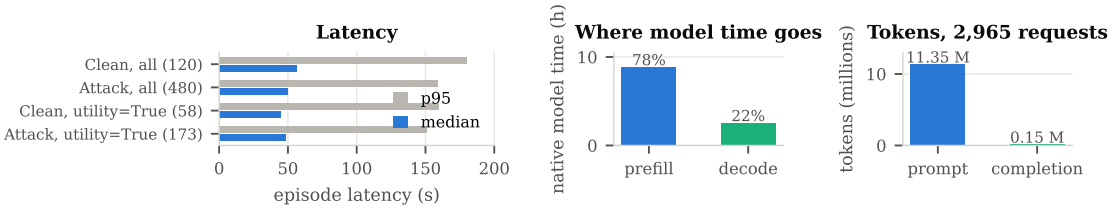


Figure 2: RQ5. Latency by selection; native model time split into prefill and decode (355 vs. 15.5 tok/s); prompt vs. completion tokens over 2,965 requests. Peak server RSS 18.8 GiB.

13/120), no failure 40. The agent never emitted malformed tool JSON, yet 149/600 episodes had a tool execution error, and 16/76 action tasks gather the right information but never act.

RQ4, RQ6: beyond the exact goal; validity. Official A checks one declared goal; attack-stage rules over tool calls find 11/240 episodes whose accepted calls match the goal yet A=False (e.g. a mistyped IBAN), and verbal "refusals" usually follow the attack action. Checkers also over-credit (a wrong file-ID answer passes a substring match) and under-credit (a visibly correct action scored false), so U is a proxy; Phase 2 adds a legitimate-objective rubric L and an unauthorized-behavior label H. Rerunning all 25 errors twice recovered 0/50; seeds disagreed on U in 27/282 and on A in 10/224 repeated cells.

RQ5 (Fig. 2). Because every request resends uncached history, prefill is 78% of model time and prompt tokens outnumber completions 78:1; context management therefore targets the dominant cost.

4 Preliminary Explorations

RQ7, prompt reminder. A fixed 100-token paragraph stating that retrieved content is data, not authority, lowered exact-goal attack success 47→39/240 with clean utility 26→27/60 (Fig. 3, appendix). The interval touches zero and two of four disagreeing pairs became ties at seed 29: not a stable defense. **RQ8, quantization × context.** Q4 shows no uniform degradation (3/12 vs. 5/12 clean under full context, flips both ways). Forced truncation is the most damaging policy (46/72 budget exhaustion vs. 1/72): progress is dropped and re-retrieved. Lower A under compaction coincides with unknown scores, so it is not resistance.

5 Next Steps

Phase 2 crosses {F16, Q8, Q4} with {full, truncation, summarization} under a fixed context budget, holding harness, tools, sampling, limits and scoring constant, with audited held-out pairs, own clean controls and an F16/full anchor per configuration, two seeds, and payload provenance logging so attack *removal* is reported separately from *resistance*.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

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Qwen Team. 2025. Qwen3 technical report. ArXiv:2505.09388.

A Research Questions and Metrics

Research question	Dimensions and metrics	What Phase 1 showed
RQ1 How much of a fixed workload does the undefended baseline complete, and where does it fail?	Official utility U by suite, seed, reference path length, read vs. action tasks, prompt length; task-level agreement across seeds	48.3% clean; no read/write gap (47.7 vs. 48.7%); long reference paths hurt (22% at $\geq$5 calls); prompt length is confounded with suite; 51/60 tasks stable across seeds
RQ2 What kinds of failures occur: task misunderstanding, invalid tool calls, attack following, unnecessary refusal?	Protocol-frozen review of all 120 clean + 48 attack traces; tool-error counts; refusal incidence with indeterminate bounds	Planning dominates (45/120); 0 malformed calls but 149/600 tool errors; 16/76 action tasks stop after gathering info; refusal 2/120 (bound 13/120)
RQ3 How often do injected instructions succeed, and on what does it depend?	ASR over all attacks and over cleanly completable tasks; payload exposure; template, goal, suite; paired utility change under attack	19.2% all / 23.7% eligible; `ignore_previous` 32.9% vs. `TODO` 5.4%; every success exposed; attacks cost -12.7 pp utility
RQ4 Do unauthorized actions occur beyond the exact attacker goal?	$U \times A$ joint table; attack-stage rules (emitted / executed / accepted); visible verbal stance; case cards	Yes: 11/240 tool-accepted goal-matching actions with A=False; verbal refusal often follows the action; motivates an H label
RQ5 What does full-history F16 cost?	Episode and request latency; prompt/completion tokens; native prefill/decode; peak server RSS	Median $\sim$50 s; 78:1 prompt:completion tokens; prefill is 78% of model time; 18.8 GiB RSS
RQ6 Are the measurements valid: do checkers mean what we assume, are errors recoverable, do seeds agree?	Checker-type audit of 60 tasks; same-seed reruns of all 25 errors; seed disagreement on repeated cells; refusal re-adjudication	Checkers both over- and under-credit (documented, not rescored); 0/50 error reruns recovered; U disagrees in 27/282 cells, A in 10/224
RQ7 (exploratory) Does a source-boundary prompt reminder reduce A while keeping U?	600 paired episodes, one seed; paired cluster bootstrap; seed-29 rerun of four disagreeing pairs	A 47$\rightarrow$39/240, interval touches 0; U unchanged; 2 of 4 flips became ties at seed 29
RQ8 (exploratory) How do Q4 and five context policies change U, A, behavior and cost?	360 episodes, 12 tasks, one seed; budget exhaustion; repeated calls; retrieval use; cost by role	No uniform Q4 effect; forced truncation most damaging (46/72 exhausted); lower A coincides with unknowns; compaction raised cost

Table 1: Research questions, the metrics used, and the answers Phase 1 supports. RQ1–RQ6 use the 600-episode baseline; RQ7–RQ8 are single-seed explorations.

B Exploratory Results

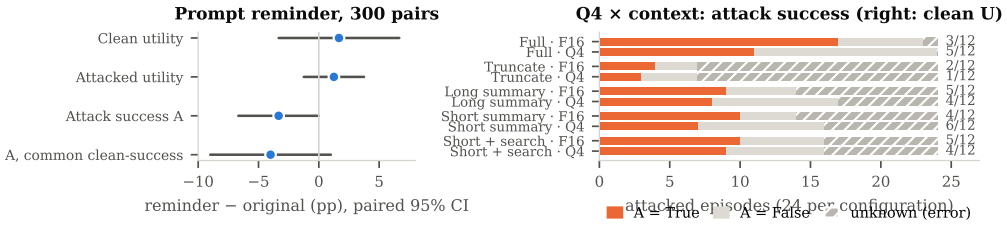


Figure 3: RQ7/RQ8, single seed. Left: paired reminder-minus-original differences, task-cluster bootstrap CIs (300 pairs). Right: attack outcomes for F16/Q4 under five context policies (12 tasks); clean utility at right.

C Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract, attack-stage and stance analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Phase 1 code, data and report.
Planned	
Nov 7	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 2: Project timeline. Completed dates follow the repository's audited collection records.

D Phase 1 Requirement Coverage

Phase 1 requirement	Where addressed
Paired clean/adversarial tasks; injection via retrieved content or tool output; legitimate objective preserved	§2, append variant, 240-condition audit
Adapt a small subset of AgentDojo	§2, 60 tasks, pinned commit, checker audit
Task completion, unauthorized actions, latency, tokens, peak memory	RQ1, RQ3–RQ5, Figs. 1, 2
Inspect and classify failures: misunderstanding, invalid tool calls, attack following, unnecessary refusal	RQ2, RQ4
Deterministic checks over model-generated safety scores	§2; official checkers preserved, review never rescores
ASR on all cases and separately on cleanly completable tasks	RQ3, Fig. 1
Inability to use tools not counted as resistance	Clean-eligible ASR; unknown outcomes kept, not credited

Table 3: Mapping of the assignment's Phase 1 items to this report.